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The Persistence and Determinants of Current Account Deficit of FYROM: An Empirical Analysis

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Abstract

This paper intends to analyze the empirical linkage between current account deficits and a wide range of macroeconomic indicators. Based on the recent econometric techniques we investigate the potential determinants of current account positions in FYROM for both, the short run and long run dynamics covering the period 1998q1-2013q4. By applying the ARDL approach, we estimate the regression equation and present the model with variables that best explain current account developments. Empirical results indicate that there is strong support for co integration relationship between current account, fiscal balance, financial development, terms of trade and trade openness. The financial development, fiscal balance and terms of trade are positively correlated with the current account balance, while openness to international trade is negatively correlated with the current account balance. Moreover, the inclusion of lagged current account as endogenous variable in the model indicates the current account deficits persistence.

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1. Introduction

The current account deficit is one of the main indicators of external imbalance of global economies. In the recent years the continuous growth of global imbalances is in the center of debate among economists and policymakers. Actually, global imbalances refer to the large deficits and surpluses of the current account positions in the global economy. Persistent global imbalances can be seen as continuous financing of net negative consumption of deficient economies by net savings of surplus economies. The global financial crisis of 2008 shed light on the importance of relations between states. As a consequence, it was raised debates if the current global imbalances pose a significant factor to the financial crisis spreading internationally or not? Europe has attracted much attention in this direction since until then the economic policies of the European Central Bank (ECB) have succeeded, particularly in the monetary policy by keeping inflation low and stable, as well as the sustainable fiscal policies. However, the crisis changed circumstances, since all member states of the European Union (EU) broke the budget constraints, while some of them are on the eve of a sovereign debt crisis.

Financial crises revealed the need for a comprehensive thoughtful of the main factors affecting the current account developments in the economy. Moreover, the latter is considered as one of the most useful analytical indicators of external imbalances that should attract more attention in the empirical studies and economic policies simultaneously. Regardless the relatively extensive theoretical literature on this topic, only a limited number of studies exist that empirically examine the impact of macroeconomic variables on the current account. The lack of empirical evidence is of apprehension, given that the current account balance is a key indicator of future economic performance. However, in recent years, the development of a number of theoretical inter-temporal models demonstrates the importance of the revealed determinants of current account. The empirical applications of these models are designed on the basis of national accounts identity, which treats the current account balance as the difference between national savings and investment.

As the most of the transition countries, FYROM is facing by a large current account deficit for a long period of time. During the last two decades and particularly after 1998, the country has registered a permanent deficit of the current account in the balance of payments which reached as high as -12.8 of GDP in 2008. This situation can bring serious implications for the national economy and the future perspective of the country. Thus, it is of particular interest to be identified the indicators that determine mostly the current account deficits. For this purpose the study intends to analyze the empirical linkage between current account balance and a broad set of macroeconomic variables for the case of FYROM. Based on the econometric approach we investigate the determinants of current account positions for both, the short run and long run dynamics covering the period 1998q1-2013q4.

The rest of the paper is structured as follows: the second section explores the literature review on empirical findings of current account balance; the third section illustrates the overall economic performance and current account developments of FYROM. In the fourth section, we briefly explain the methodology, empirical approach and data that are used for the empirical results. The fifth section explores the empirical findings, while in the last section the conclusions of the study are given.

2. Literature review

The issue regarding the determinants of current account balance attracted scholars' interest in the early 1980s. There are some empirical studies that analyze the determinants of current account, to ascertain the level of the current account that can be considered as 'regular' for a state, based on a set of macroeconomic and structural attributes. These economic fundamentals usually include GDP per capita, demography, fiscal balance and initial net foreign assets (NFA). The main studies on short-term current account fluctuations are based on the assumption that current account acts as a buffer against temporary shocks to income, to smooth consumption and to maximize welfare. The most prominent studies to this direction have been conducted by Ghosh (1995), Ghosh and Ostry (1995), Glick and Rogoff (1995), Nason and Rogers (2006) and Kraay and Ventura (2000).

Another group of studies are focused on medium and long-term determinants of current account from an empirical perspective. Initially, Dabelle and Faruquee (1996) and later the Consultative Group on the exchange rate (CGER) by IMF (2006) empirically have analyzed the structural determinants of the current account balance of

developed countries from a saving-investment viewpoint. Calderon et al. (2002, 2007), Chinn and Prasad (2003), Chinn and Ito (2007) and Lee et al. (2008) extended their analysis including developing countries.

For instance, Debelles and Faruquee (1996) use a panel of 21 developing countries over the period 1971-'93 and a wide array of cross-section data of 34 developed and developing countries. The purpose of their study was to examine the factors that may affect the long-term current account determination. Short-term variables are used to ascertain that the position of the current account is not necessarily in the long term equilibrium. Their empirical findings suggest that the relative income and demographics have a considerable impact on the current account balance (in the most of cases). They also find that the fiscal policy has a significant influence on the current account. Further, they come across that the short-term variables such as changes in the real exchange rate, business cycle and changes in the terms of trade have also significant impact on the current account developments.

Calderon, Chong and Loayza (1999) complement the work of Debelles and Faruquee (1996), by applying the latest econometric techniques, to observe whether the respective variables are jointly endogenous to the current account deficit. Also, by conducting a simple econometric model, they separately assess the temporary and permanent relationship of current account deficit with its determinants. Thus, they estimate the determinants of the current account balance of 44 developing countries during 1966-1995. Amongst others, their findings suggest: current account deficits are moderately persistent; GDP growth worsens the current account deficit; temporary increase of public savings or private savings improves the current account balance; temporary appreciation of the real exchange rate or the deterioration of terms of trade wide the current account deficit but their impact in the long term is not significant; a temporary increase of the level of economic growth in developed countries leads to a reduction of current account deficits in developing countries (in the short and long term); world high interest rates in developed countries, reduce the deficit of current account in developing countries.

Chinn and Prasad (2003) take up quite different econometric methods compare to Calderon, Chong and Loayza (1999) and their attention is to the medium-term determination of current account. Their empirical analysis consist of a wide database of 18 developed countries and 71 developing economies, investigating for the medium term fluctuations of current account. Findings show that the budget balance, the initial net foreign assets (NFA) and financial indicators are positively correlated with the current account balance in developing countries. Their model was further expanded by Gruber and Kamin (2007), analyzing the impact of the financial crisis in the current account balance. Chin and Ito (2008) update the work of Chinn and Prasad (2003), including the identified factors by Bernanke (2005), respectively the impact of legal and institutional indicators. According to them, these factors may also be important determinants of savings and investment decisions, as they affect the rate of return of these activities.

Recent empirical studies have examined the determinants of current account based on specific characteristics of the state. Calderon et al. (2007) focus on developing economies and low income states. Among others, they found that the appreciation of the real exchange rate (which is in accordance with the Mundell-Fleming model) and deterioration of total terms of trade (which is compatible with the effect Harberger-Laursen-Metzler, HLM) deteriorate the current account deficit.

Medina, Pratt and Thomas (2010) study the determinants of current account balance for developing countries. Their results suggest that the fiscal balance significantly affects the current account, as well as they argue that the increase of net foreign assets improves the current account balance.

Hermann and Jochem (2005) and Bussiere et al. (2004) examine the excessive deficits of the current account in the most of new member states of EU. Based on panel data estimations deficits in these countries are mainly determined by the relative income per capita and high capital investments. The difference of incomes between new and old member states is really shrinking, while the effect of budget deficits is quite small, since they are mainly financed by private savings. Analyzing the developing countries of Europe, Rahman (2008) finds that foreign direct investment and transition index have negative and significant impact on the current account balance, while remittances have considerable positive influence. Zanghieri (2004) based on a simple theoretical model (with budget constraint) finds that the financial deepening and budget deficit have significant impact on current account. Moreover, FDI inflows play an important role for financing the external deficit. Prudential fiscal policy is essential in order to maintain external debt at manageable levels.

Herman and Winker (2008) investigate the current account imbalances in the context of convergence of income and analyze it comparing the developing countries in Asia and developing countries in Europe. They claim that the

development of the financial market and financial integration are important factors in determining the current account balances. More developed financial markets and higher financial integration are associated with higher deficits / lower surplus in the current account. This is possible because during the process of convergence, countries with more developed financial markets and integrated are able to borrow from abroad and thus to increase domestic consumption and savings. However, some indicators of financial development and integration not fully explain the different models of real convergence and the current account in developing Europe and developing Asia.

The literature on the determinants of current account for oil exporting states is very limited. Morsy (2009) highlights that the factors that determine the current account balance, in addition to fiscal balance and the balance of fuels, are also sources of oil, the age dependency ratio and increased oil production. Arezki and Hasanov (2009) conclude that the fiscal policy has a greater impact on the current account balance for the oil exporting states comparing to the other economies.

Aristovnik (2006) examines the main determinants of the current account balance to assess the possible extension of current account deficits in selected transition economies, Eastern Europe and the former Soviet Union. The analysis covers the period 1992-2003 and using different estimates for both empirical models (A and B) the author find: a) Economic growth has negative effect on the current account balance, implying that the increase of growth rate is associated with higher rise of investments than savings; b) Confirm the hypothesis of stages of development, as the poorer countries in the region register higher deficits in the current account; c) Fiscal balance has positive and significant impact on the current account, confirming the validity of the hypothesis of a double deficit; d) The appreciation of the real exchange rate and the deterioration of terms of trade (TOT) aggravate the current account deficit the current account deficits in transition economies. According to the empirical model A, FYROM for the period 2000-2003 captures a current account deficit of 7.1% of GDP, while according to model B, the current account deficit is lower 5.1% of GDP.

3. Economic development of FYROM

FYROM experienced a difficult and protracted transition to a market economy due to various external shocks and internal problems. Pre-transition GDP levels were not restored until the middle of the previous decade because robust GDP growth was delayed. Its economy is characterized by relatively slow growth (see Table 1). Realized rates of economic growth are not satisfactory in terms of strengthening the country's economy and raising the standards of living of the population. The highest growth rate of 6.1% was recorded in 2007. While several indicators have shown significant improvement, the global financial crisis has affected the economy. Regarding other indicators, low inflation is maintained through prudential monetary policy (fixing the domestic currency against the euro), while unemployment is still high (29% in 2013). Unlike some more advanced transition countries, the country did not experience large capital inflows from the developed economies. Also, developments in the external sector show higher deficits in the current account over the critical value of '5% of GDP', which points out the problem of competitiveness. Especially, the persistence of high trade deficit is a serious weakness and suggests that the structural factors weigh on competitiveness.

Table 1. Selected macroeconomic indicators, average for periods.

	1995-2003	2003-2013	1995-2013
Real GDP growth (in %)	1.4	3.05	2.55
Inflation (average)	4.1	2.6	3.04
Unemployment rate	33.5	31.5	31.4
Budget balance (central government and funds, in % of GDP)	-1.6	-2.6	-2.7

Source: NBRM; Author's calculation

FYROM is a member of the World Trade Organization (WTO) since April 2003. The membership in this organization implied the need of adjusting its trading rules with WTO rules. Starting from 2006, the country takes an important step towards integration into the EU, signed the Central European Free Trade Agreement (CEFTA).

Regarding the balance of payments position during the transition period, the FYROM's economy liberalized several items, where the current account flows were liberalized in 1998 year. From this year onwards the country has recorded a current account deficit in its balance of payments. The current account deficit has fluctuated between 0.4 and 12.8% of GDP. The lowest deficit of the current account was recorded in 2006 year, by 0.4% of GDP, which was largely determined by the substantial growth of private transfers. Since 2007, the situation began to change significantly, where the current account recorded the highest deficit ever in 2008 by 12.8% of GDP. Current account deterioration was caused by the increase in the trade deficit, partly as a result of deteriorating terms of trade; decrease of private transfers, probably due to the impact of events in Kosovo, local elections etc; lower factorial net income (Telecom of FYROM paid dividends 80 million); the rise of global prices of food and energy. All these factors led to higher nominal imports and on the other hand the slow global growth reduced exports.

By the end of 2009, the tightening of the current account deficit of 6 percentage points of GDP was as a result of lower trade deficit. The deficit of trade of goods has fluctuated within the levels of 10.8 and 26.2% of GDP. Most of the trade deficit is covered by high inflows of private transfers, averaging over 75% for the period 1998-2012, dominated by remittances from abroad workers. They allow the current account deficit to be affordable (average 4.5% for the period 1998-2012). However, despite their positive contribution to the current account, private transfers are a component around which there is a great uncertainty[†].

4. Methodology and data

The determinants of current account balance of FYROM are firstly analyzed by various model specifications estimated by OLS method, in view of the time series stationary properties. Having into consideration, that the analyzed series weren't with the same order of integration, we further employed more sophisticated econometric approach for testing the cointegration among variables, such that the lagged dependent variable in an autoregressive distributed lag (ARDL) model. The authors, Pesaran and Shin, (1997); Pesaran et al. (2001), reveal in their studies the advantages of the ARDL modeling approach. The main advantage is its applicability regardless of the order of integration of the series (the series can be either $I(0)$ or $I(1)$). Thus, it is not necessary pre-testing the stationary of the variables, which differs from the other methods where series should be of the same order or in most cases of the first order $I(1)$ as a circumstance for cointegration testing. Additionally, the ARDL method gives better estimates for small samples as compared with the FM-OLS and it takes enough numbers of lags through the process in a general-to-specific modeling framework, which captures the data generating process. Also, in case when all variables are $I(1)$, then for calculating the long run parameters, it is not necessary to increase the number of the regressors in order to correct simultaneously the residual autocorrelation and the problem with the endogenous variables. Furthermore, the dynamic error correction model - ECM can be derived from ARDL through a simple linear transformation. So, through this approach we study the long run relationship and dynamic interactions among the variables that best determine the current account position. Also, the short run dynamics have been estimated using ARDL approach to cointegration.

Following the recent techniques in the econometric analyses, and in conformity with the sample size, which consists of 64 observations (1998Q1-2013Q4) and the different order of integration of the included variables, it was evident that ARDL modeling approach is the most appropriate procedure for estimating the determinants of current account balance, for the case of FYROM.

[†] Based on a survey conducted in 2007 (Roberts et al. 2008) , only 38.6% of remittances were sent through formal channels, while in most cases, 44.4% of abroad workers transported physically their money. Therefore, more than half of money transfers which come as private transfers are not registered.

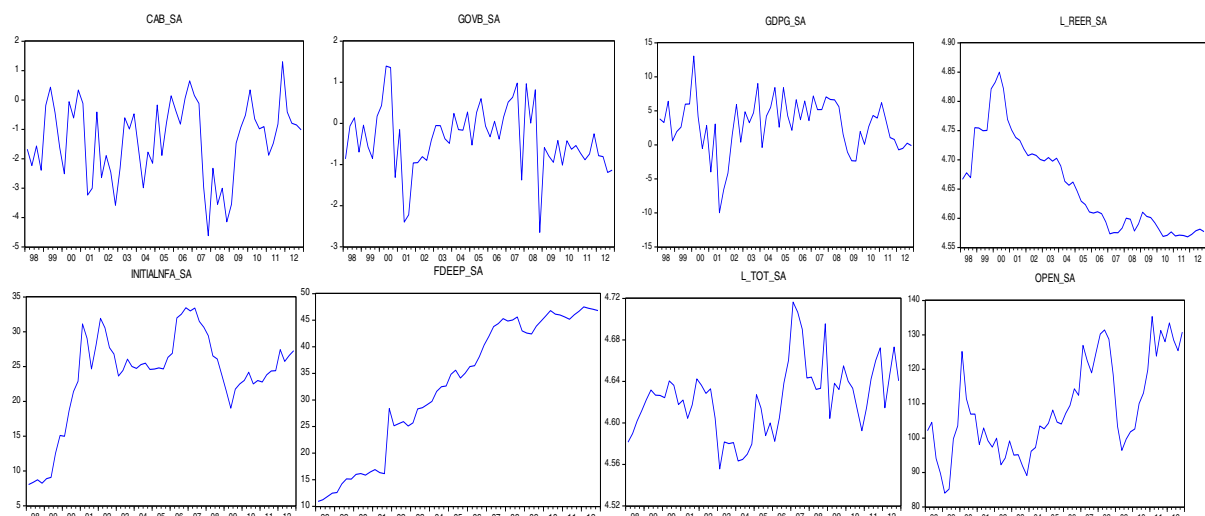
The ARDL approach consists of two steps. The first step is to examine the existence of long run relationship among all the variables in an equation and the second step is to estimate the long run and short run coefficients of the same equation. We run the second step only if we find a cointegration relationship in the first step. This step determines the appropriate lag lengths for the independent variables. The general form of the error correction version of the ARDL model is presented as:

$$\Delta Y_t = \alpha_0 + \sum_{i=1}^{p-1} \alpha_i \Delta Y_{t-i} + \sum_{j=0}^{p-1} \alpha_j \Delta X_{t-j} + \beta_1 Y_{t-1} + \beta_2 X_{t-1} + \varepsilon_t$$

where Δ represents change, Y_t is the dependent variable, X_t represents the explanatory variable and ε_t the error term. The first part of the equation with α coefficients, represents the short run dynamics of the model, while the second part, with β coefficients, represents the long run relationship between the dependent and explanatory variable. The second step is testing the cointegrating relationship using F-test. Specifically, the hypotheses $H_0 = \beta_1 = \beta_2 = 0$ means that there is no cointegration or long run relationship between the dependent and explanatory variables, while the rejection of H_0 means that at least one of the long run coefficient is different of null, which implies of long run relationship between at least one explanatory and dependent variable. Pesaran and Pesaran (1997); Pesaran et al.(2001) computed two sets of asymptotic critical values for testing cointegration. The first set assumes variables to be $I(0)$, the lower bound critical value and the other $I(1)$, upper bound critical value. If the F-statistic is above the upper bound critical value, the null hypothesis of no cointegration can be rejected irrespective of the orders of integration of times series. On the contrary, if the test falls below the lower bound critical value, the null hypothesis cannot be rejected. Finally, if the statistic falls between these two sets of critical values, the result is inconclusive. The second phase is selecting the appropriate ARDL model (the optimal number of lag lengths) and calculating the long run coefficients.

4.1. The data

The study considers 9 important potential determinants of the current account balance which enable the researchers to decide which of them mostly determine the current account deficits of FYROM. The quarterly data are used in the empirical analysis, covering the period 1998q1-2013q4. The main sources of data are the National Bank (NBRM) and the State Statistical Office (SSO). The data for the fiscal balance come from the Ministry of Finance (MF), whereas the oil prices are provided by IMF. The most of the data are used as percent of GDP, while the rest as indexes. All the used series previously are adjusted for the effect of seasonality using ARIMA X12. The used variables in the analysis are presented in (Figure 1) below, whereas the data description in (Table 2).



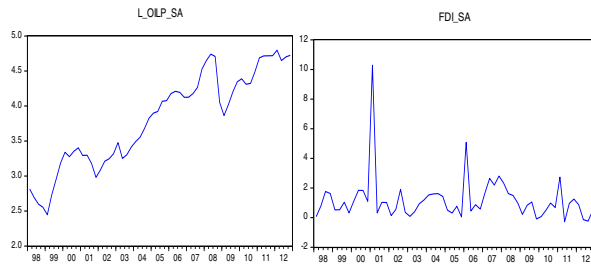


Figure 1. Used variables in the empirical analysis

Sources: National Bank and State Statistical Office of FYROM, MF and IMF

Table 2. Description of data and data sources

Variable	Abbreviation	Description	Source
Current account balance	CAB	The current account balance (% of GDP)	NBRM, SSO
Budget balance	GOVB	The central government budget balance (% of GDP)	MF, SSO
Economic growth	GDPG	The growth rates of real GDP (%)	SSO
Exchange rate	REER	Real effective exchange rate, base year 2006=100	NBRM
Net foreign assets	NFA	Net foreign assets of the overall banking system (in % of GDP)	NBRM, SSO
Financial development	FDEEP	The monetary aggregate M2 (in % of GDP) as a proxy of financial development	NBRM, SSO
Terms of trade	TOT	Changes in the relative prices of exports and imports, base year 2005 = 100	NBRM calculations
Openness	OPEN	Ratio of exports plus imports to GDP	NBRM, SSO
Oil prices	OILP	Oil prices for barrel in \$	IMF
Foreign Direct Investments	FDI	Foreign direct investments, net (in % of GDP)	NBRM, SSO

Table 3. Descriptive statistics for the variables

	CAB	GOVB	GDPG	REER	NFA	FDEEP	TOT	OPEN	OILP	FDI
Mean	-1.351	-0.409	2.98	105.11	24.141	33.106	101.88	109.066	-1.886	1.168
Median	-1.123	-0.461	3.351	101.42	25.028	35.556	101.97	105.873	-1.737	0.693
Maximum	2.002	1.772	14.37	128.08	36.989	47.326	112.75	134.114	-0.428	10.45
Minimum	-5.887	-3.443	-10.19	95.92	8.124	10.408	94.96	81.179	-3.96	-0.552
Std. Dev.	1.608	0.978	3.907	8.564	6.632	12.605	3.758	13.858	0.741	1.498
Skewness	-0.483	-0.319	-0.457	0.87	-0.933	-0.453	0.549	0.241	-0.633	4.237
Kurtosis	3.189	4.005	4.723	2.851	3.649	1.727	3.384	1.985	3.164	25.61
Jarque-Bera	2.507	3.664	9.824	7.886	10.09	6.307	3.498	3.264	4.211	150.6
Observ.	64	64	64	64	64	64	64	64	64	64

Source: Authors' calculation

5. Empirical results

The analysis first starts with time series properties of the variables checked through Augmented Dickey-Fuller (ADF) and Philips-Perron unit root testing procedures. The results indicate that series of current account balance (CAB), government balance (GOVB), FDI, and GDP growth are stationary in their levels and integrated of order I(0), whereas regarding the other series appear to contain a unit root in their levels but become stationary in the first difference. The series of FDEEP and OILP become stationary once de trended. The test results are summarized in Table 4 below.

Table 4. Unit root test statistics of series

Variables	Test	Level		First Difference		
		t-statistics	p-value	t-statistics	p-value	Decision
CAB	ADF	-3.107**	0.025	-	-	I(0)
	PP	-6.072***	0.000	-	-	
GOVB	ADF	-2.970*	0.053	-	-	I(0)
	PP	-3.091**	0.048	-	-	
GDPG	ADF	-2.914*	0.055	-	-	I(0)
	PP	-2.977*	0.050	-	-	
REER	ADF	-1.157	0.738	-3.800***	0.002	I(1)
	PP	-1.346	0.711	-5.102***	0.000	
NFA	ADF	-1.872	0.632	-3.924***	0.003	I(1)
	PP	-1.926	0.418	-7.412***	0.000	
FDEEP	ADF	-1.107	0.426	-3.234**	0.032	I(1)
	PP	-1.968	0.324	-4.551***	0.000	
FDI	ADF	-3.540**	0.013	-	-	I(0)
	PP	-3.991***	0.008	-	-	
OPEN	ADF	-0.198	0.618	-3.100**	0.042	I(1)
	PP	0.397	0.781	-3.102**	0.040	
OILP	ADF	-2.245	0.190	-3.924***	0.003	I(1)
	PP	-2.926	0.118	-7.412***	0.000	
TOT	ADF	-1.467	0.550	-3.639***	0.005	I(1)
	PP	-1.522	0.487	-3.878***	0.002	

The notifications ***, ** and * denote 1%, 5% and 10% level of significance, respectively.

Authors' calculations

The different order of integration of variables reveals that it is appropriate to use the Pesaran et al. (2001) procedure for estimating the empirical linkage between current account balance and its determinants. However, before displaying the results of the ARDL model, in the following table are presented a variety of model specifications estimated by OLS. As explanatory variables, which include almost any specification, are the current account with a time lag (CAB_{t-1}), the financial development (FDEEP) and terms of trade (TOT). The obtained results show that the coefficients of these variables are statistically significant in all model specifications, while trade openness and government balance show statistical significance in the most of model specifications. Regarding the other analyzed variables don't show any significant effect on current account development.

The coefficient of lagged CAB is positive and statistically significant which shows the current account deficit persistence. Also, the results suggest that the coefficient of financial development is positive and statistically significant that is in line with other studies (see for example Edwards, 1995). The linkage between terms of trade (TOT) and CAB is also positive (HLM effect) and statistically significant. Opening to international trade negatively affect the current account balance. The result of the coefficient of fiscal balance show the validity of the hypothesis of a double deficit (twin deficits hypothesis), while the connection between the fiscal balance and the current account balance is positive and statistically significant. The third specification adds the real effective exchange rate (REER), as a typical indicator of competitiveness. The negative coefficient obtained for the REER means that the appreciation of the domestic currency reduces exports, leading to a deterioration of the current account balance (in

accordance with elasticity approach, Mundell-Fleming model and absorption approach). However the relationship between REER and CAB is statistically insignificant. In the fourth specification we add the variable of economic growth as one of the potential determinants of current account balance. Our goal is to observe whether the real growth rates of GDP worsen the current account balance (for example through increased consumption). The results show that this indicator is also statistically insignificant, although the negative sign is consistent with the theory. Further, as the majority of empirical studies in the fifth specification we make efforts to monitor the effect of net foreign assets (initial) on the current account. Results (coefficient obtained for NFA) suggest that higher influx of NFA's would improve the current account balance, but the impact of NFA's indicator on the current account balance is not significant. The next specification includes foreign direct investment (FDI) but the coefficient doesn't show statistical significance. The last specification, intends to control the impact of energy prices. The sign of the coefficient obtained for this indicator is in line with the theory, but is not statistically significant. Thus the variables in the second model of (Table 5) was selected as the determinants of current account balance that best explain the current account developments: current account with a time lag, the financial development, the terms of trade, fiscal balance, and trade openness.

Table 5. Results of different specifications estimated with OLS

Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
CAB_SA(-1)	0.520 (5.10)***	0.583 (5.63)***	0.565 (5.40)***	0.571 (5.48)***	0.499 (4.22)***	0.601 (5.74)***	0.577 (5.63)***
D_FDEEP_SA	0.209 (1.92)***	0.214 (3.03)***	0.202 (2.84)***	0.196 (2.71)***	0.188 (2.61)***	0.218 (3.09)***	0.223 (3.19)***
D_L_TOT_SA	11.323 (2.22)***	11.318 (2.32)***	11.253 (2.31)***	11.601 (2.38)***	10.879 (2.25)***	11.112 (2.28)***	13.876 (2.71)***
D_OPEN_SA		-0.039 (-2.00)**	-0.039 (-2.03)**	-0.029 (-1.35)	-0.034 (-1.77)**	-0.041 (-2.11)**	-0.054 (-2.49)**
GOVB_SA		0.291 (1.81)**	0.297 (1.85)**	0.383 (2.09)**	0.241 (1.48)	0.322 (1.97)**	0.219 (1.32)
D_L_REER_SA			-7.748 (-1.14)				
GDPG_SA				-0.042 (-1.05)			
D_NFA_SA					0.100 (1.42)		
FDI_SA						-0.094 (-1.08)	
D_L_OILP_SA							1.62 (1.49)
R2		0.42	0.50	0.51	0.51	0.51	0.52

D- denotes the first difference and SA-seasonally adjustment of variables. ***, ** and * denote 1%, 5% and 10% level of significance, respectively.

Author's calculations

5.1. ARDL results

The next analysis investigates which of previously found determinants of current account balance affects its development in the short run and which in the long run using the ARDL model. The ARDL specification of potential current account balance determinants is represented in the following equation.

$$\Delta CAB_t = \alpha_0 + \alpha_{1i} \sum_{i=1}^n \Delta CAB_{t-i} + \alpha_{2i} \sum_{i=0}^n \Delta FDEEP_{t-i} + \alpha_{3i} \sum_{i=0}^n \Delta TOT_{t-i} + \alpha_{4i} \sum_{i=0}^n \Delta GOVB_{t-i} + \alpha_{5i} \sum_{i=0}^n \Delta OPEN_{t-i} + \beta_1 CAB_{t-1} + \beta_2 FDEEP_{t-1} + \beta_3 TOT_{t-1} + \beta_4 GOVB_{t-1} + \beta_5 OPEN_{t-1} + \varepsilon_t$$

This equation was estimated in two stages. In the first stage of the ARDL procedure, the long run relationship was established in two steps. First, a bounds F-test was applied to the equation to test the joint significance of β coefficients for lagged variables. Second, the order of lags on the first-differenced variables was obtained from unrestricted VAR by means of the Akaike Information Criterion (AIC) and Schwarz Bayesian Criterion (SBC). The results of the bounds F testing are displayed in Table 6.

Table 6. Results for the bounds test for cointegration

K	N	F statistic	Upper (1%)	Lower (1%)	Upper (5%)	Lower (5%)	Decision
1	4	6.43	2.52	3.77	2.38	4.12	Cointgration

k-is the lag length, n-is the number of variables in the equation other than lags of dependent variable.

Authors' calculation

SBC selects the optimal lag length as one, although AIC selects four. There is no autocorrelation in the residuals for the first three lags. Therefore, optimal lag length was selected as one. The results of the test according to the one lag model indicate that the null hypothesis of no long-run relationship is rejected with both 5 and 1 percent significance levels since the F statistic is above the upper bound levels.

The next step is to determine the optimal lag length for ARDL model. Based on the AIC the lag lengths of the model for long run equilibrium were determined as ARDL (4, 2, 1, 0, 0), represented by the following equation.

$$CAB_t = \alpha_0 + \alpha_{1i} \sum_{i=1}^4 CAB_{t-i} + \alpha_{2i} \sum_{i=0}^2 FDEEP_{t-i} + \alpha_{3i} \sum_{i=0}^1 TOT_{t-i} + \alpha_{4i} \sum_{i=0}^0 GOVB_{t-i} + \alpha_{5i} \sum_{i=0}^0 OPEN_{t-i} + \varepsilon_t$$

While the lag lengths based on the AIC for the short run equilibrium was determined as ARDL (3, 2, 0, 3, 0), or:

$$\Delta CAB_t = \alpha_0 + \alpha_{1i} \sum_{i=1}^3 \Delta CAB_{t-i} + \alpha_{2i} \sum_{i=0}^2 \Delta FDEEP_{t-i} + \alpha_{3i} \sum_{i=0}^0 \Delta TOT_{t-i} + \alpha_{4i} \sum_{i=0}^3 \Delta GOVB_{t-i} + \alpha_{5i} \sum_{i=0}^0 \Delta OPEN_{t-i} + \delta_1 ECT_{t-1} + \varepsilon_t$$

The empirical results of the long run coefficients are shown in Table 7 below. All estimated coefficients are found to be statistically significant and with same signs as previously estimated by OLS. The coefficient of financial development (FDEEP) is statistically significant at 5% level. The fiscal balance (GOVB) is strongly significant and positively correlated with current account deficit. Also, the coefficient of terms of trade (TOT) is strongly significant, while the coefficient of trade openness is significant at 10% level. The coefficient of determination R^2 is 0.59 indicating that minimum 59% of changes in the current account balance are explained by the selected variables. Thus, results confirm that these variables determine the position of current account deficit of FYROM in the long run. Additionally, the diagnostic tests of serial correlation, functional form, normality and heteroscedasticity were performed. The results show that the long run model, in the most of cases, passes through all diagnostic tests.

Table 7. Long run coefficients based on ARDL (4, 2, 1, 0, 0)

Variable	Coef.	Std.Err.	t	P> t
CAB(-1)_sa	0.3899073***	0.1218731	3.20	0.003
CAB(-2)_sa	0.0688493	0.1210025	0.57	0.572
CAB(-3)_sa	-0.2678386**	0.1245695	-2.15	0.037
CAB(-4)_sa	0.3056731**	0.1240922	2.46	0.018
FDEEP_sa	0.0828304**	0.0418677	1.98	0.042
FDEEP(-1)_sa	-0.2440148**	0.1129112	-2.16	0.036
FDEEP(-2)_sa	0.1735185*	0.0923005	1.88	0.067
TOT_sa	0.1979767***	0.0626658	3.16	0.003
TOT(-1)_sa	-0.1539701**	0.0637910	-2.41	0.020
GOVB_sa	0.6906908***	0.2017245	3.42	0.001
OPEN_sa	-0.0278418*	0.0204530	-1.56	0.080
_cons	-2.2146121	5.9443680	-0.37	0.711

D- denotes the first difference and SA-seasonally adjustment of variables. ***, ** and * denote 1%, 5% and 10% level of significance, respectively.

Author's calculations

Table 8. Short run coefficients based on ARDL (3, 2, 0, 3, 0)

Variables	Coef.	Std. Err.	T	P> t
D_CAB(-1)_sa	-0.0257247	0.2045431	-0.13	0.901
D_CAB(-2)_sa	0.0591339	0.1425488	0.41	0.680
D_CAB(-3)_sa	-0.2557871*	0.1376236	-1.86	0.070
D_FDEEP_sa	0.1132634*	0.1671933	1.69	0.093
D_FDEEP(-1)_sa	-0.3046205***	0.0903978	-3.37	0.002
D_FDEEP(-2)_sa	-0.1827048*	0.0927762	-1.97	0.056
D_TOT_sa	0.1868416***	0.0497048	3.76	0.001
D_GOVBSa	0.8127645***	0.1875943	4.33	0.000
D_GOVBS(-1)_sa	0.4322334*	0.2630604	1.64	0.102
D_GOVBS(-2)_sa	0.0679316	0.2294713	0.30	0.769
D_GOVBS(-3)_sa	-0.3156117*	0.1892856	-1.67	0.103
D_OPEN_sa	-0.0482801*	0.0252990	-1.91	0.063
ECT(-1)	-0.5202206**	0.2327873	-2.23	0.031
_cons	0.2929377	0.1992827	1.47	0.149

D- denotes the first difference and SA-seasonally adjustment of variables. ***, ** and * denote 1%, 5% and 10% level of significance, respectively.

Author's calculations

The results of the short run coefficient estimates associated with the long run relationships obtained from the ECM version of ARDL are presented in Table 8. As it takes time for these variables to exhibit their impact on the current account balance, we expect the later to be related to both, of that period and lagged values of the selected variables. This means that the current account balance is a function of or depends on current as well as lagged values

of financial development, terms of trade, fiscal balance and trade openness. The results of this analysis also found strong support for the short run relationship between the current account and these determinants. All coefficients of variables are statistically significant, except the coefficients of first and second lags of current account balance and the second lag of government balance. Also the signs vary among the coefficients of lagged variables of selected variables. The coefficient of the error correction term (ECT_{t-1}) measures the speed of adjustment of variables to restore to equilibrium in the dynamic model. It should have a statistically significant coefficient with negative sign. In this model it appears with negative and statistically significant at 5% level, ensuring that the long run equilibrium can be attained. A highly significant error correction term confirms the existence of a stable long run relationship among variables (Bannerjee et al. 1998). The coefficient of the lagged error correction term (ECT_{t-1}) is -0.52, this means that after any shock, it takes two periods for the current account balance to return to its long run equilibrium level.

6. Conclusions

The issue of current account deficits remains one of the most debatable topics which ever more upset macroeconomic policymakers. In the center of the debates have been questions about the determinants of the current account deficit, which may vary from country to country, depending on the country's specific economic characteristics. The current account balance of FYROM has been continually in deficit, while escalating deficit periods include the political crisis of year 2001 and the global financial crisis of 2008. The later ended with a record of current account deficit of 12.8%. The current account performance of the country is determined mainly by the trade balance and the persistence of high trade deficit as a significant weakness. As a soothing factor of trade deficit are high and sustainable private transfers. The foreign trade structure shows a high concentration of exports, which are closely related to terms of trade. In these circumstances, assessment of the factors that determine the current account balance is very useful for the country. Moreover, it is necessary for policy makers to be aware of the current account response to these foundations, in order to develop appropriate policies to reduce the current account deficit. Therefore, this research intends to investigate the determinants that affect mostly the performance of current account balance of FYROM for the period 1998Q1-2013Q4. By applying the ARDL approach, we estimate the long run and short run determinants of current account deficit. Empirical results indicate that there is strong support for co integration relationship between current account balance, fiscal balance, financial development, terms of trade and trade openness. The results suggest that the present developments of current account balance are strongly determined by past developments. The relationship between financial development (monetary aggregate M2) and the current account is positive and statistically significant, meaning that the growth of M2 has the effect of improving the current account balance; increased fiscal deficit causes an increase in the current account deficit (so-called double deficit hypothesis); the positive relationship of the current account and trade conditions, suggest that the improvement in the terms of trade leads to a reduction of the current account deficit; opening to international trade deteriorates current account deficit and this is expected because a higher degree of openness can lead a country to pursue ongoing deficits.

Observing the external sector position of FYROM, and the profile of the current account, especially during the global financial crisis, our analysis suggests that structural reforms are necessary and competitiveness of exports should be improved. Also, given that higher financial development means more savings this will have a positive effect on the current account. Therefore, we advocate that promoting saving measures are vital for improving the current account balance performance.

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